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SAFETY PLAN GENERATOR

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Of course. As STRUCTMIND AI, I will generate a comprehensive lifting, rigging, and erection safety plan based on your project parameters and the requirements of OSHA 1926 Subpart R.

Steel Erection Safety Plan

Project: 3-Storey Steel Structure Erection Location: [Project Site Address] Date: [Date] Prepared By: STRUCTMIND AI (For Site Competent Person Review)

Disclaimer: This document is a template and must be reviewed, modified, and approved by a qualified and competent person for the specific site conditions, personnel, and equipment involved in this project. All activities must comply with the manufacturer's recommendations and all applicable federal, state, and local regulations, including OSHA 1926 Subpart R.

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1. Lifting Plan Notes

This section details the parameters for crane operations, focusing on the critical picks.

• Heaviest Pick: Spandrel Girder

• Weight: 4.2 tons (8,400 lbs).

• Rigging Weight (Estimated): 400 lbs (spreader bar, slings, shackles).

• Total Load (Gross): 8,800 lbs.

• Critical Pick Parameters:

• Crane Working Radius (Max): 70 ft.

• Load Dimensions: [Enter Girder Length x Width x Height].

• Crane Sizing & Selection:

• Crane Type (Proposed): 90-110 Ton All-Terrain or Hydraulic Truck Crane (Example). The final selection must be based on the specific crane's load chart.

• Load Chart Verification: The selected crane's load chart must show a capacity of at least 12,000 lbs at a 70 ft radius with the specified boom length and configuration. This provides a conservative safety margin over the 8,800 lb gross load.

• The operator must have the crane's load chart available at all times. The operator will not exceed 75% of the crane's rated capacity for any pick.

• Pre-Lift Requirements:

• Ground Conditions: Crane setup area must be inspected by a competent person. Ground must be firm, level, and graded. Crane mats (timber or composite) shall be used under outriggers to distribute the load if ground compaction is not certified to withstand the outrigger point loads.

• Pre-Lift Meeting: A daily pre-lift meeting will be held with the crane operator, signal person, riggers, and ironworkers involved. The meeting will cover the day's picks, communication methods, and potential hazards.

• Inspections: The crane will have a current annual inspection certificate. The operator will perform a documented daily inspection before the first lift.

• Weather: Lifts will be suspended if wind speeds exceed 25 mph (or a lower speed as specified by the crane manufacturer). All lifting operations will cease in the event of an electrical storm. An anemometer must be on-site.

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2. Rigging Plan (For 4.2-ton Spandrel Girder)

This plan details the specific rigging assembly for the heaviest pick. All rigging gear must have legible capacity tags and be inspected daily by a qualified rigger.

• Rigging Method: Two-point vertical pick using a spreader bar to prevent sling stress and member buckling.

• Rigging Components & Specifications:

• Spreader Bar (1):

• Type: Manufactured Spreader Bar.

• Required Capacity (WLL): Minimum 5 tons (10,000 lbs).

• Top Shackle (1):

• Type: Bolt-Type Anchor Shackle.

• Required Capacity (WLL): Minimum 6.5 tons (13,000 lbs) to connect the crane hook to the spreader bar.

• Slings (2):

• Type: Wire Rope Slings with softeners or Synthetic Web Slings.

• Required Capacity (WLL): Each sling must have a vertical hitch capacity of at least 2.5 tons (5,000 lbs). This accounts for each sling supporting half the load (4,200 lbs) with a safety margin.

• Lower Shackles (2):

• Type: Screw-Pin or Bolt-Type Shackles.

• Required Capacity (WLL): Minimum 3.25 tons (6,500 lbs) each, to connect slings to the girder's designated lifting points.

• Rigging Procedure:

1. The qualified rigger will inspect all components before assembly.

2. Attach the top shackle to the crane hook and the spreader bar's center lifting eye. Ensure the shackle pin is properly secured.

3. Attach the two slings to the spreader bar ends using the lower-rated shackles.

4. Attach the slings to the girder's designated lifting points. Use softeners if wire rope slings are used on painted surfaces or sharp edges.

5. Use two (2) taglines, attached at opposite ends of the girder, to control the load during the swing and placement.

6. The signal person will have a clear line of sight to the load and the operator at all times.

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3. Fall Protection Plan (OSHA 1926.760)

100% fall protection is mandatory for all personnel working at heights of 15 feet or more.

• Primary Fall Protection System: A Personal Fall Arrest System (PFAS) is the primary method.

• Components: Full-body harness, energy-absorbing lanyard or self-retracting lifeline (SRL), and a suitable anchor point.

• Anchor Points: Must be capable of supporting 5,000 lbs per person attached. Approved anchor points include structural members, engineered horizontal lifelines, and beam clamps specified for fall arrest. Do not anchor to joists or decking.

• Inspection: Each worker will inspect their PFAS daily before use.

• Connectors: Per OSHA 1926.760(b), connectors working between 15 and 30 feet (or two stories, whichever is less) may work without being tied off only while actively engaged in connecting and must have a PFAS available to tie off immediately after connecting. Above 30 feet, connectors must be tied off at all times.

• Controlled Decking Zone (CDZ): A CDZ may be established for leading-edge decking work.

• Use: Only authorized personnel trained in CDZ procedures are allowed inside.

• Boundaries: The CDZ will be clearly marked with control lines no more than 90 feet wide and 90 feet deep from the leading edge.

• Fall Protection: Workers in a CDZ must be protected from fall hazards of more than two stories or 30 feet.

• Holes and Openings: All floor openings and open-sided floors will be protected by guardrails, covers, or safety nets as soon as decking is installed.

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4. Erection Stability & Bracing Plan

This plan ensures the structure remains stable throughout the erection process. All activities must follow the sequence outlined in the project's erection drawings.

• Foundation and Anchor Rods (1926.755):

• The controlling contractor must provide written notification that the concrete footings have cured to the strength specified in the design documents.

• All column base plates must be secured with a minimum of four (4) anchor rods. The erector will verify anchor rod placement and condition before setting columns.

• Erection Sequence & Bracing:

1. The first two columns must be set, plumbed, and stabilized with temporary guys or bracing cables before any other members are connected.

2. Temporary bracing must remain in place until permanent stability is achieved through the installation of floor beams, diagonal bracing, and fully secured metal decking as specified in the erection drawings.

3. Two-Bolt Rule (1926.756(a)): Before the crane load is released from any member, it must be secured with at least two bolts per connection, drawn up wrench-tight.

4. The erection sequence will proceed bay by bay, ensuring each completed bay is fully plumbed, braced, and stable before moving to the next.

• Load Placement: Do not place construction loads (e.g., bundles of decking, equipment) on unbraced steel. Loads must be placed over primary support members only after the structure's stability is confirmed by the competent person.

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5. Hazard Register with Controls

Hazard	Potential Harm	Control Measures
Crane Overturning	Crushing, Multiple Fatalities	Adherence to load charts, firm ground/matting, certified operator, daily inspections, wind speed monitoring.

Hazard	Potential Harm	Control Measures
Dropped Load	Struck-by, Crushing	Qualified rigger, daily rigging inspection, use of taglines, established "no-go" exclusion zone under loads.
Falls from Height	Severe Injury, Fatality	100% tie-off >15 ft, use of PFAS, guardrails, hole covers, connector-specific training.
Structural Collapse	Multiple Fatalities, Crushing	Strict adherence to erection sequence, installation of temporary bracing, verification of foundation strength, 2-bolt rule before unhooking.
Swinging Load / Struck-by	Impact Injuries, Fractures	Use of taglines, clear communication between operator and signal person, controlled access zone.
Pinch Points	Hand/Limb Amputation	Hands-free tools where possible, hand placement awareness training, clear communication during connections.
Adverse Weather	Dropped Load, Crane Instability, Electrocution	On-site anemometer, suspension of work at 25 mph winds (or lower), immediate cessation of work during lightning.

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6. PPE & Site Access Notes

• Minimum Required PPE for All Personnel:

• Hard Hat (ANSI Z89.1)

• Safety Glasses with Side Shields (ANSI Z87.1)

• High-Visibility Vest or Shirt

• Steel-Toed Work Boots

• Task-Specific PPE:

• Ironworkers/Riggers: Leather gloves, Personal Fall Arrest System (PFAS).

• Grinding/Welding: Face shield, welding hood, appropriate protective clothing.

• High Noise Areas: Hearing protection (earplugs or earmuffs).

• Site Access & Control:

• Erection Zone: A controlled access zone will be established around the crane's swing radius and directly under the erection area. Only essential personnel (erection crew, signal person, inspector) are permitted inside this zone.

• Signage: The area will be clearly marked with "DANGER - OVERHEAD WORK" and "AUTHORIZED PERSONNEL ONLY" signs.

• Laydown Yard: The steel laydown area will be organized to allow safe access. Steel members will be stored on dunnage and chocked to prevent rolling.

• Daily Briefing: All site personnel will attend a daily safety briefing to review the plan, tasks, and associated hazards for the day.